Data & Software Engineering Case Study

U.S. Rail Freight Activity Monitoring Platform

Build a platform that ingests public U.S. freight rail data and surfaces trends through a dashboard.

The user is an equity researcher who wants to understand what is moving through the U.S. freight rail network, what is driving changes, and whether service conditions are improving or deteriorating. This is primarily an engineering exercise; any forecasting or machine learning is a bonus.

Please write the backend in Python, but all other architecture choices are up to you. Use any tools that help you work effectively. Limit your total time spent on the project to five hours.

Constraints

- No account signups, API keys, OAuth credentials, or paid services
- All data must come from publicly accessible sources
- Must work from a fresh clone; do not rely on pre-loaded data artifacts committed to the repo

Data Sources

Use the following public rail datasets. Read the linked dataset pages and metadata before building, then choose a reproducible way to acquire and refresh the data with support for at least five years of history rather than a hand-picked sample.

Source	URL	Role
Rail Carloadings by Commodity	https://agtransport.usda.gov/Rail/Rail-Carloadings/tb7g-kn5i	Primary weekly freight activity dataset
STB Railroad Service Metrics	https://agtransport.usda.gov/Rail/STB-Railroad-Service-Metrics/axkm-yjzy	Weekly service-performance dataset
Surface Transportation Board Rail Service Data	https://www.stb.gov/reports-data/rail-service-data/	Source documentation / reference

Feel free to go above and beyond if you see opportunity.

What to Deliver

Alongside the complete source code, include a concise write-up that covers:

- Your approach and methodology
- The main challenges you ran into and how you addressed them
- Any assumptions you made along the way
- A few sentences on how you imagine an analyst using the system
- A self-assessment of your submission — where it's strong, where it falls short, and what you'd improve with more time. Be as quantitative as you can

Before you begin, complete the case study compliance form.

When you are finished, push your work to a **private GitHub repository** and share it with the following users:

- jg-joshtennefoss
- jg-reidhartman
- jg-nickirwin